

National College of Ireland
Higher degree diploma in Science in Data Analytics
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Statistics 1
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1.Introduction

1.1 Background information

Retailers usually use discounts on their items to let consumers make more purchases, but the real effect of discount on revenue is not very clear.

1.2 Statistical Question

Do discounts significantly increase revenue(sales)?

1.3 Summary of the report

This report uses descriptive statistics and hypothesis testing to determine whether discounts considerably increase sales.

2.Literature Review

Several studies have found that promotions like giving discounts influence consumer purchasing behavior.

The research paper of Priyadharshini and Rajesh Kumar examined discount and promotional offers on purchasing decisions [1]. They found that discounts significantly encourage consumer buying and increase short-term sales, but continuous promotions can also have a negative impact on customer loyalty. Another finding is that personalized and targeted discounts are more effective in customer satisfaction. This supports the idea that discounted transactions may create higher revenue than non-discounted transactions.

Another study of Kanchan Shekhawat et al from MIT World Peace University also explored how various marketing techniques affect consumer buying behavior [2]. They found that different types of promotion influence market demand, and that customers' perception of product price depends on the type of discount offered. The results showed that there was a significant moderate effect on customers' feeling of product risk and preference on price reduction by giving discounts. This shows that discounts can stimulate market demand, and this aligns with the statistical question of whether discounted items make higher sales.

Furthermore, in Nishanth and Sivasakthi research, they analyze how discounts impact shopping styles, consumer purchases, and loyalty [3]. The results reveal that promotional offers do increase consumer spending, and strategic and consistent promotions are vital in boosting consumer satisfaction and sales. These observations represent that discount strategies can directly grow sales performance and support the hypothesis tested in this report.

Additionally, Sudhaker and colleagues studied the influence of promotional activities on consumer behavior in retail settings [4]. The findings showed an obvious relationship between consumer sales promotion and consumer purchasing behavior. The researchers concluded that price discount is broadly applied by retailers presenting their goods in Tema, Ghana. Statistically, the study significantly showed that discounts affect consumer behavior and sales performance.

3.Methodology

3.1 Data Description

The dataset used in this report is from Kaggle <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>. This dataset contains information on product sales, including variables such as discount and sales revenue. The discount column includes whether transactions were reduced by discount, and the sales column represents the revenue that was generated by each transaction. These columns are used to compare discounts and without discounts sales performance. Descriptive statistics showed that the sales variable was with a long, right narrow tail skewness, and the mean value is much higher than the median value. That indicates that the dataset has extremely high-value transactions.

3.2 Data Preparation

Before implementing any analysis, the dataset executed some cleaning processes such as checking missing values and incomplete rows were removed. This report will present descriptive statistics and one inferential statistic test. When creating some descriptive visualization, especially the sales variable, a long, narrow tail skewness was shown. To reduce the skewness, a log transformation was applied and then created a new variable `log_sales`. This will help with

consistent variance and is suitable for applying the t-test statistics. To assess the normality of log_sales by Q-Q plot, additional columns should be added to the dataset: a rank column (takes the small to large of log_sales) and a theoretical quantiles column (take the value of the normal distribution at each rank position). A two groups variable was created to be used in the inferential test, such as discount and no discount values.

<i>Sales</i>		<i>Log_sales</i>	
Mean	227.5929938	Mean	4.108821796
Standard Error	5.817110485	Standard Error	0.016433896
Median	54.432	Median	3.996951827
Mode	12.96	Mode	2.561867691
Standard Deviation	581.536509	Standard Deviation	1.64289652
Sample Variance	338184.7112	Sample Variance	2.699108976
Kurtosis	183.1421646	Kurtosis	-0.551690493
Skewness	10.25396911	Skewness	0.185021309
Range	17499.506	Range	10.58188402
Minimum	0.444	Minimum	-0.811930717
Maximum	17499.95	Maximum	9.769953303
Sum	2274564.38	Sum	41063.56503
Count	9994	Count	9994

3.3 Statistical Methods and Justification

The visual assessment made by Q-Q plot comparing the observed values of log_sales with theoretical quantiles to make sure that the dependent variable is approximately normally distributed for the t-test. This project aims to compare the average log_sales between two independent variables- discount and no discount transactions. Therefore, a two-sample unpaired t-test was selected. To decide on the correct t-test formula, the f-test was executed to determine the equality of variances (equal or unequal variances) between two groups. An unpaired sample t-test is suitable to evaluate a statistically significant difference in the mean of log_sales between two groups that are not related and directly aligned to the question “Do discount significantly increase revenue?”.

4. Results

4.1 Descriptive Statistics

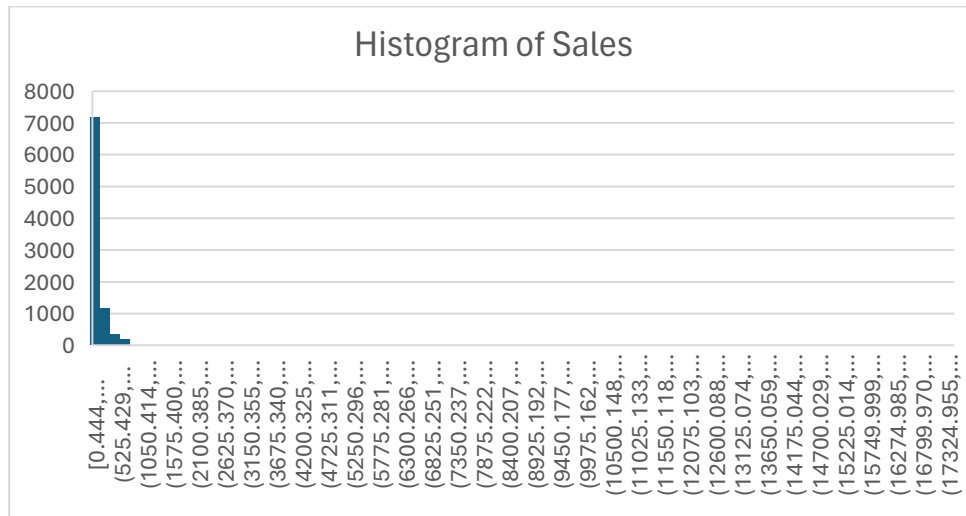


Figure 1. Distribution of raw sales values showing strong right-skewness.

The above figure shows that the raw sales variable has long right tail skewness, indicating that there are a few extreme product prices included in the sales transactions. Since, this skew is not suitable for t-test assumptions of normalized data distribution, a log transformation should be added to become a stabilized variance and enhance normality.

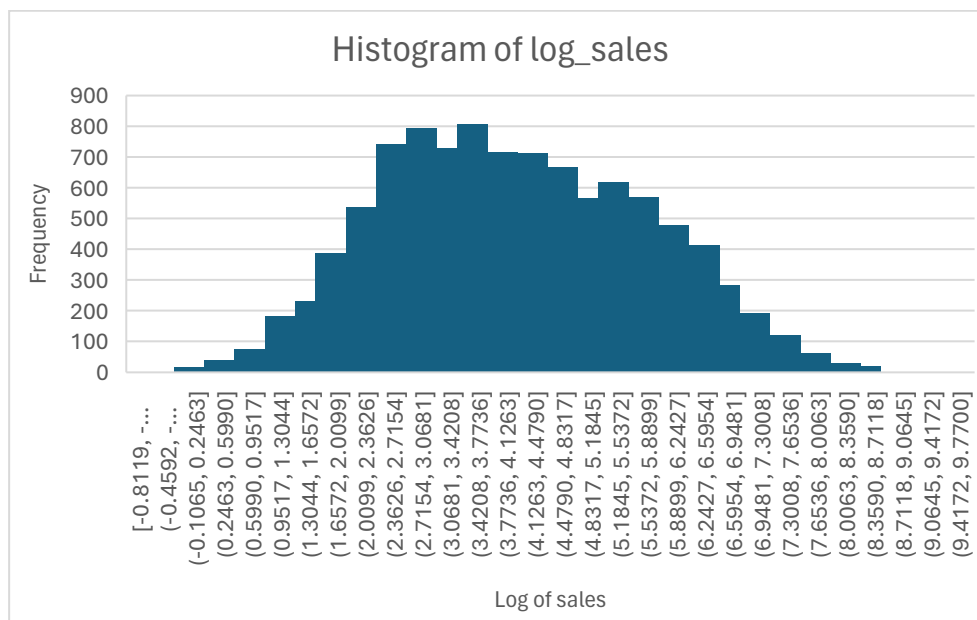


Figure 2. Distribution of raw sales values showing strong right-skewness.

After applying the log transformation, the distribution of log_sales becomes more normalized. This transformation ensures that the values in log_sales can be used as independent sample data.

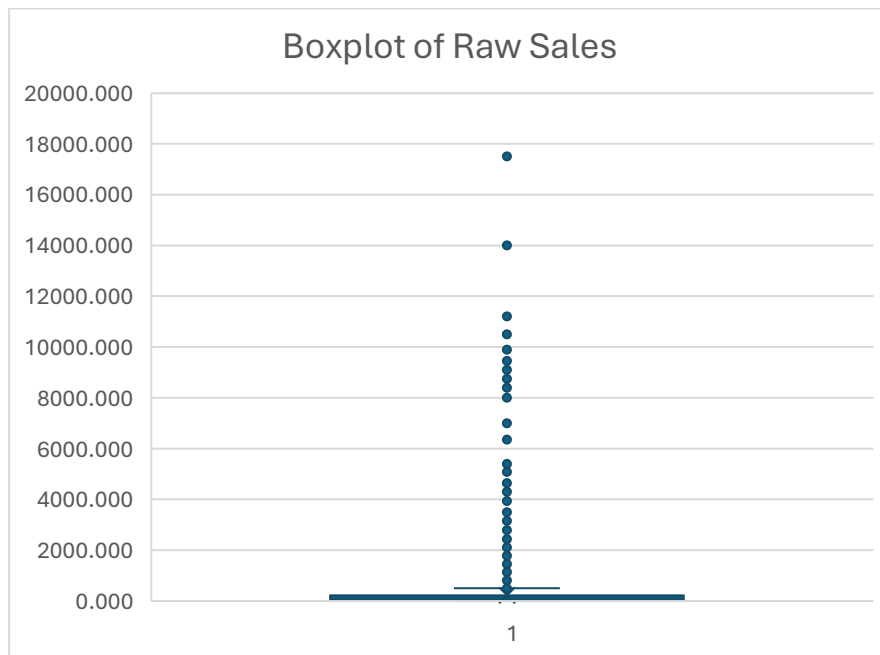


Figure 3. Boxplot of raw sales values showing extreme right-skewness and high-value outliers.

The boxplot of raw sales variable shows that there are a long upper whisker and a few extremely high-value outliers. This indicates that the sales data is skewed and cannot apply to the t-test.

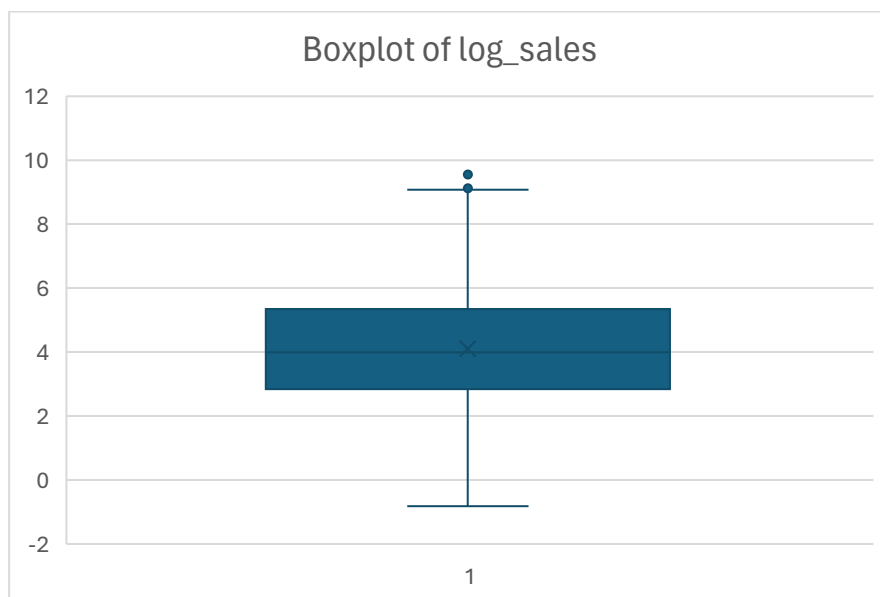


Figure 4. Boxplot of log_sales after log transformation.

After the log transformation has been implemented, the boxplot is now more symmetric, and the effect of outliers is minimized. This means that the transformation has been effective at reducing skew and ensuing to use as a sample dataset for unpaired samples t-test.

4.2 Normality Check

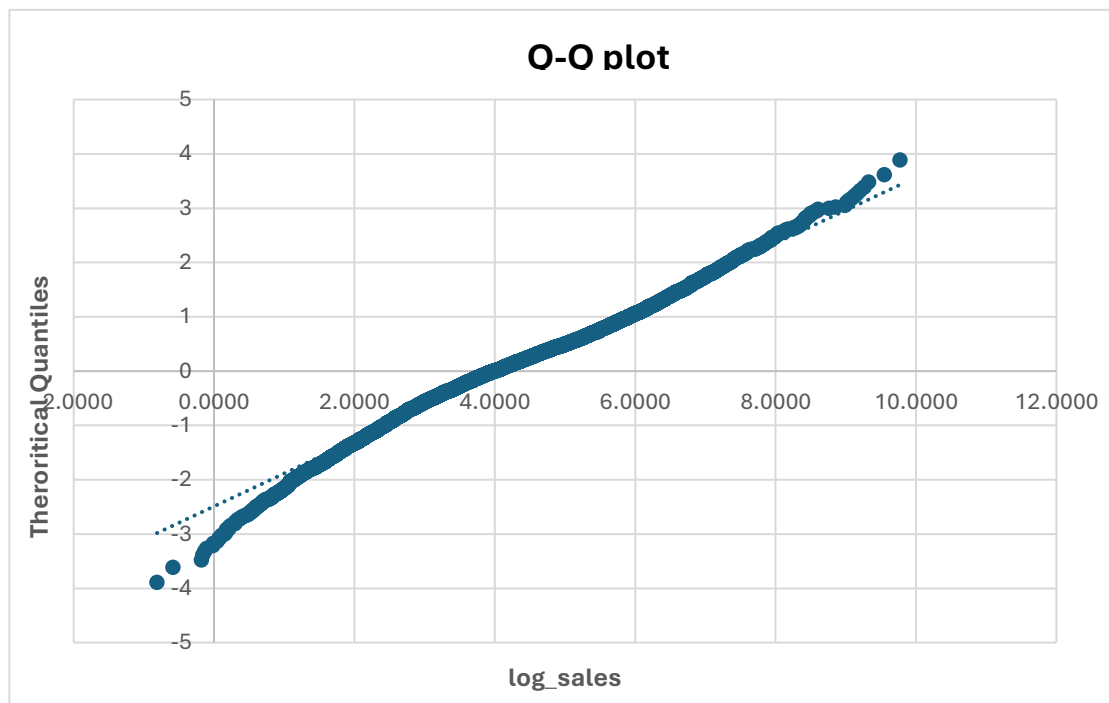


Figure 5. Q-Q plot comparing observed log_sales values with theoretical normal quantiles.

The above figure shows that the log-transformed sales data is normally distributed. This supports that the log_sales value is normalized, and the t-test can be applied.

4.3 Group Statistics

Discount Group	Mean	Sum	Count	SD	Max	Min	Var
Discount	5.41052969	28113.1123	5196	1.03114864	9.7699533	3.88740239	1.06326752
No Discount	2.69913563	12950.4528	4798	0.80490264	3.88740239	0.81193072	0.64786826
Total	4.1088218	41063.565	9994	1.64281432	9.7699533	0.81193072	2.6988389

Figure 6. Comparison of mean log_sales for discounted and non-discounted transactions.

The mean log sales of discounted transactions was 5.41 and the mean log sales of non-discounted transactions was 2.7. This big difference was because of some discounted transactions, including a much higher value in sales revenue than non-discounted transactions.

4.4 Inferential Statistics

F-Test Two-Sample for Variances

	Discount Log_sales	No Discount Log_sales
Mean	5.410529689	2.699135633
Variance	1.063472188	0.648003321
Observations	5196	4798
df	5195	4797
F	1.641152372	
P(F<=f) one-tail	1.01246E-67	
F Critical one-tail	1.047711826	

Before solving the t-test, the f-test was applied to determine whether the variances in the log sales of the discount and no discount group are the same or not. The hypotheses were established as below:

Null hypothesis (H0)

The two groups have equal variances. $H_0 : \sigma^2_{discount} = \sigma^2_{no\ discount}$

Alternative hypothesis (H1)

The two groups have unequal variances. $H_1 : \sigma^2_{discount} \neq \sigma^2_{no\ discount}$

The test resulted in a f value of 1.64 and a very low p-value of 1.01246E-67 because of a huge dataset. This extremely small p value shows that the difference between the variance of the two groups is unlikely by chance. Another examination is that f statistics value (1.64) is greater than f critical value (1.05). Therefore, we can reject the null hypothesis with 95% confidence. As a result, the unequal-variances version of the t-test (Welch's t-test) was chosen.

t-Test: Two-Sample Assuming Unequal Variances

	<i>Discount</i> <i>Log_sales</i>	<i>No Discount</i> <i>Log_sales</i>
Mean	5.410529689	2.699135633
Variance	1.063472188	0.648003321
Observations	5196	4798
Hypothesized Mean Difference	0	
df	9727	
t Stat	147.1047799	
P(T<=t) one-tail	0	
t Critical one-tail	1.645010296	
P(T<=t) two-tail	0	
t Critical two-tail	1.9602079	

The independent samples t-test is used to determine whether there is a statistically significant difference between the mean log_sales of two groups. The hypotheses were established as below:

Null hypothesis (H0)

Discounted transactions do not have higher mean log_sales than non-discounted transactions.

$H_0: \mu_{discount} \leq \mu_{no\ discount}$

Alternative hypothesis (H1)

Discounted transactions do have higher mean log_sales than non-discounted transactions.

$H_1: \mu_{discount} > \mu_{no\ discount}$

Formula of the t-Statistics

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

X1 is the mean log_sales for discounted group

X2 is the mean log_sales for no discount group

s^2 is the variance of each Group

n_1, n_2 are the number of participants in each Group

The result of $t = 147.1$

Degrees of Freedom (df)

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

The result of $df = 9727$

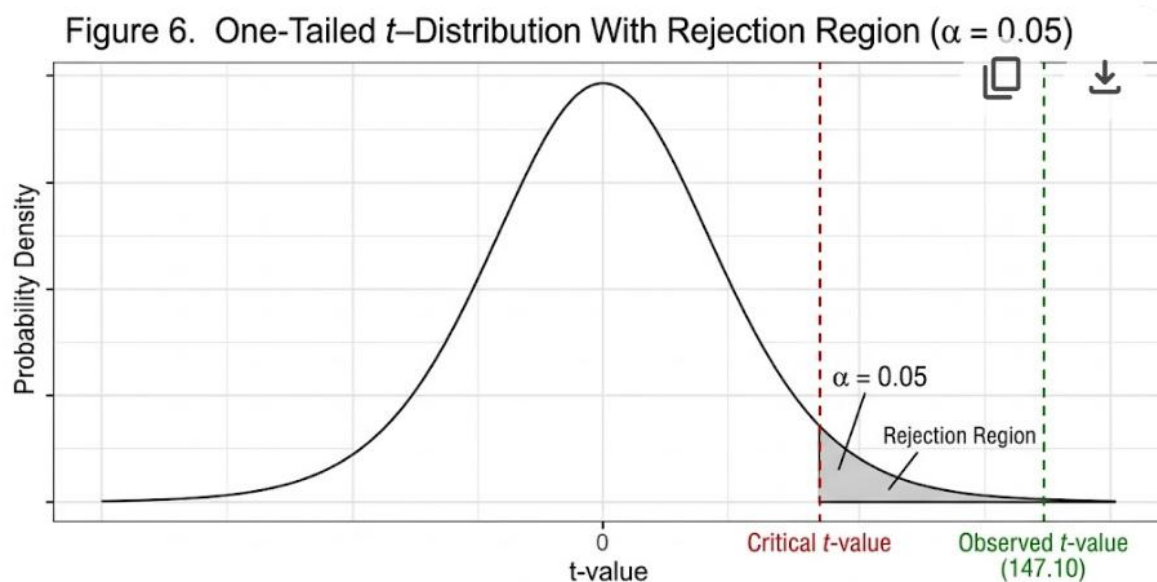


Figure 6. One-Tailed t -Distribution With Rejection Region ($\alpha = 0.05$) with $df = 9727$.

According to p-value interpretation, both one-tailed and two-tailed p-values are extremely small ($p=0$). Therefore, $p < 0.05$, reject the null hypothesis.

According to t statistics value interpretation, the one-tailed t-test got the result $t=147.1$, which is far greater than the t critical value ($t_{critical}= 1.64$). As shown in figure 6, the t value falls inside the rejection region. Therefore, we can confidently reject the null hypothesis and strongly accept the alternative hypothesis of discounted transactions have higher mean log_sales than non-discounted transactions. Which means that discounts do significantly increase revenue.

4. Conclusion

In this report, the data was just grouped into two variables of discount and no discount, even though the superstore offered different discount levels. Another hypothesis test could be used to solve that problem. However, according to the inferential statistics, the discount effects significantly increase revenue for the retail business.

References

- [1] Ms. Priyadharshini S, Dr. Rajesh Kumar K, "IMPACT OF DISCOUNTS AND OFFERS ON CONSUMER PURCHASING," *International Journal of Scientific Research and Engineering Development*, p. 5, 2025.
- [2] Kanchan Shekhawat, Adarsh Puri, Neha Sharma, Harshad Wagh, Priyanka Kokatnur , "A STUDY ON THE IMPACT OF DISCOUNTS ON CONSUMER BUYING BEHAVIOUR," *International Journal of creative research thoughts (IJCRT)*, p. 5, 2020.
- [3] Mr. S. NISHANTH ¹, Ms. D. SIVASAKTHI ², "A STUDY ON IMPACT OF DISCOUNTS AND OFFERS ON CONSUMER BUYING BEHAVIOUR WITH SPECIAL REFERENCE TO D-MART IN COIMBATORE CITY," *International Journal of Research Publication and Reviews*, p. 3, 2025.
- [4] 1P. Sudhaker,2Shruthi Teegala,3Sudhakar Madhavedi,4Ch.Prathima, "THE INFLUENCE OF PROMOTIONAL ACTIVITIES ON CONSUMER BEHAVIOR IN RETAIL SETTINGS A STUDY ANALYZING PURCHASE DECISIONS, BRAND PERCEPTIONS, AND SALES IMPACT," *NeuroQuantology*, p. 11, 2020.